

Pre-Reads: Data Structures – Lists



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Pre-Reads

Foundation

Recommended

Resource



Associated Lectures (1)



Description



Discussions



Pre-Read: Lists – Python’s Most Useful Container



Why This Session Matters

Until now, your variables could hold only **one** piece of information. But real life is never about just one thing.

A shopping cart has many items. A playlist has many songs. A classroom has many students. A to-do app has many tasks.

Python needs a way to store **multiple** pieces of data together in one place. That superpower is called a **List**.

Lists are one of the most important tools you’ll ever use in Python. Once you learn them, your programs will instantly become more flexible, more powerful, and more “real-world.”



What Is a List?

Think of a List like a **stretchable shelf** where you can keep items in order. You can add things, remove things, reorder them, or look inside whenever you want.

A List can store:

- Names
- Numbers
- Favorite movies
- Grocery items
- Even other lists

It’s a container that grows and shrinks as needed. And you control everything inside it.

How We Work With Lists in Real Life

Here are simple scenarios that behave exactly like Python Lists:

A playlist

You can add a new song, remove one you no longer like, rearrange them, or shuffle them.

A to-do list

You write tasks, check some off, remove completed ones, and reorder based on priority.

A queue at a shop

People come in, people leave, people move around.

Python Lists work the same way, just digitally.

Why Lists Are So Important

With Lists, you will be able to:

Store many values together

Like all quiz scores, all students, all products.

Access anything instantly

Like asking, “What is the first item?” or “What is the last one?”

Grab a portion

Like saying, “Show me only the middle three names.”

Modify on the go

Add, remove, sort, reverse, update methods are all possible.

Repeat actions easily

You can go through your List item by item using loops.

Create new lists quickly

List comprehensions help you build new lists from old ones in a clean, readable way.

All of this makes Lists the foundation of almost everything you’ll build later.



What You Will Explore in Class

In the session, you'll learn how to:

- Create lists
- Access elements
- Pick slices
- Add or remove items
- Loop through a list
- Build new lists from existing ones

And then you'll use all of this in a fun **To-Do List mini project** where the list is the heart of your program.



Before the Session

Think about your daily life:

- Your contacts list
- Your watchlist on OTT platforms
- Your wish list on a shopping app
- Your recipe ingredients

These are all Lists.

Come curious. Once you learn Python Lists, you'll start seeing them everywhere.



Closing Thought

“Lists are where your programs stop being tiny and start feeling truly useful.”

This session will open the door to writing programs that feel closer to the real world in a way that more organized, flexible, and dynamic.
